Estimating Urban Poverty via Satellite Image Classification of Urban Morphology: A Swin Transformer Approach for Greater Dhaka

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***Abstract*—Accurate and scalable estimation of urban poverty remains a critical challenge in developing nations, where traditional survey-based methods are costly, slow, and geographically limited. This paper presents an automated pipeline for poverty estimation in the Greater Dhaka Metropolitan Area, Bangladesh, using multispectral Sentinel-2 satellite imagery and a fine-tuned Swin Transformer model. A novel labeling methodology based on the Metal Roof Index, derived from the ratio of Short-Wave Infrared (SWIR) to Near-Infrared (NIR) spectral bands, is used to generate poverty scores for 800m grid cells across a 120km x 120km study area. A dataset of 17,771 georeferenced image patches (112x112 pixels) is constructed using Google Earth Engine and a memory-efficient Python processing pipeline. A Swin-Tiny transformer pretrained on ImageNet is fine-tuned for three-class poverty classification (Wealthy, Medium, Poor) using class-weighted cross-entropy loss to address severe class imbalance. The model achieves a best validation accuracy of 97.19% with minimal overfitting, outperforming a Swin-Base baseline trained on a smaller 8,000-patch dataset. The resulting poverty map provides spatially explicit predictions across Dhaka and its surroundings, offering a reproducible, survey-free framework for poverty monitoring in data-scarce urban environments.**

***Index Terms*—urban poverty estimation, satellite imagery, Swin Transformer, remote sensing, metal roof index, Sentinel-2, Dhaka, image classification**

I. Introduction

Urban poverty mapping is a fundamental prerequisite for effective policy intervention, resource allocation, and sustainable development planning. In rapidly urbanizing cities of the Global South, conventional approaches to poverty measurement, such as household surveys and census data, suffer from significant limitations: they are expensive to conduct, infrequent in updates, and often fail to capture fine-grained spatial heterogeneity within cities [1]. The Greater Dhaka Metropolitan Area, one of the most densely populated urban agglomerations in the world, exemplifies these challenges. Dhaka hosts a complex mosaic of affluent neighborhoods, middle-income settlements, and informal slums (locally called "bastis"), often in close geographic proximity [2].

Recent advances in remote sensing and deep learning have opened new possibilities for scalable, cost-effective poverty estimation from satellite imagery. Convolutional neural networks (CNNs) have demonstrated success in extracting poverty-relevant features from aerial and satellite images [3]. Vision Transformers, and in particular the Swin Transformer architecture, have further improved upon CNNs by modeling long-range spatial dependencies through shifted window self-attention, making them especially suitable for complex scene understanding in satellite imagery [4].

- A fully automated, end-to-end pipeline from Google Earth Engine (GEE) satellite export to labeled patch generation, requiring no manual annotation.
- A novel application of the Metal Roof Index (SWIR/NIR ratio) as a proxy poverty label, derived from Sentinel-2 spectral bands, validated against ground truth via Google Earth visual inspection.
- A curated dataset of 17,771 georeferenced 112×112 pixel RGB patches covering the Greater Dhaka region, with three poverty class labels.
- A fine-tuned Swin-Tiny transformer achieving 97.19% validation accuracy with a favorable model complexity-to-dataset-size ratio, suitable for publication-grade remote sensing research.
- A spatially explicit predicted poverty map of Dhaka generated from model inference over the full dataset.

II. Related Work

A. Remote Sensing for Poverty Estimation

The use of satellite imagery for socioeconomic inference has gained substantial traction. Jean et al. [3] demonstrated that nighttime light intensity from DMSP-OLS imagery, combined with daytime imagery features extracted via transfer learning, could predict wealth indices in sub-Saharan Africa with high accuracy. Subsequent work expanded these approaches to higher-resolution imagery sources such as Landsat and Sentinel-2, enabling finer-scale poverty mapping [5].

Rooftop material classification has emerged as a particularly robust proxy for poverty in informal settlements, given that corrugated metal or tin roofs are strongly associated with low-income housing in South and Southeast Asia [6]. Spectral

indices derived from the Short-Wave Infrared (SWIR) band of Sentinel-2 have been shown to discriminate metal roofing from other materials, providing an automated labeling mechanism that avoids costly ground surveys [7].

B. Vision Transformers in Remote Sensing

The Swin Transformer, introduced by Liu et al. [4], employs a hierarchical architecture with shifted windows for computing self-attention, achieving strong performance on image recognition benchmarks while being computationally efficient. Remote sensing applications of Swin Transformers have demonstrated improvements over CNN-based methods in scene classification, object detection, and semantic segmentation of high-resolution satellite imagery [8]. The hierarchical feature representation of Swin models is well-suited to the multi-scale structure of urban scenes, where poverty-related morphological patterns manifest at both fine (rooftop material) and coarse (settlement density, street grid) spatial scales.

III. Study Area and Data

A. Study Area

The study area encompasses the Greater Dhaka Metropolitan Area, Bangladesh, centered on BRAC University (23.7806°N, 90.4003°E), within a 60 km buffer radius resulting in a 120×120 km coverage zone. The area spans a diverse range of urban morphologies including:

- **Core Dhaka**: Tejgaon, Moghbazar, Khilgaon, Rampura, Badda, Merul Badda, Aftabnagar
- **North**: Uttara, Turag, Mirpur, Pallabi, Tongi
- **South**: Old Dhaka, Lalbagh, Keraniganj, Shyampur
- **East**: Demra, Narayanganj, Siddhirganj
- **West**: Savar, Ashulia, Aminbazar

This geographic diversity ensures that the dataset captures the full spectrum of urban poverty conditions present in the metropolitan region.

B. Satellite Data Acquisition

Satellite imagery was acquired from the Sentinel-2 Surface Reflectance Harmonized collection (Copernicus/S2_SR_HARMONIZED) via Google Earth Engine (GEE), using 2023 composites with a cloud cover threshold of less than 20%. Sentinel-2 provides multispectral imagery at 10 m spatial resolution for visible and near-infrared bands, and 20 m resolution for SWIR bands (resampled to 10 m in GEE). The exported image (`bracu_full_image4.tif`) contains RGB (B4, B3, B2), NIR (B8), and SWIR (B11) bands, split into four tiles due to GEE export size limitations.

C. Poverty Label Generation

Poverty labels are derived from the Metal Roof Index (MRI), defined as:

$$\text{MRI} = \frac{\text{SWIR}_{B11}}{\text{NIR}_{B8}} \tag{1}$$

Pixels with MRI > 1.0 are classified as tin or corrugated metal roofing, a well-established proxy for low-income housing in the South Asian urban context [6]. For each 800×800 m grid cell overlaid on the study area, the poverty score is computed as the proportion of pixels exceeding the MRI threshold:

$$\text{poverty_score} = \frac{\sum_i \mathbf{1}[\text{MRI}_i > 1.0]}{N_{\text{pixels}}} \tag{2}$$

where N_{pixels} is the total number of valid pixels in the grid cell. A score of 0 indicates a fully wealthy area with no metal roofing, while a score of 1 indicates complete coverage by tin roofs. Labels and grid geometries were exported as `bracu_labels_roofs4.csv` from GEE.

Three ordinal poverty classes are defined from continuous poverty scores:

TABLE I: Poverty Class Definitions

Class	Poverty Score Range	Label
Wealthy	0.00 – 0.33	0
Medium	0.33 – 0.66	1
Poor	0.66 – 1.00	2

D. Dataset Construction

Satellite tiles were processed using a memory-efficient Python pipeline (`main_v3.py`) implementing the following steps:

1) **Tile merging**: Sequential tile loading and stitching using `rasterio`, processing one tile at a time to avoid memory overflow.
2) **Normalization**: Per-band normalization using the 2nd–98th percentile range, clipping satellite outliers caused by atmospheric artifacts or sensor noise.
3) **Patch extraction**: Non-overlapping 112×112 pixel patches are extracted with a sliding window; patches with mean pixel value below 5 (empty or cloud-masked regions) are discarded.
4) **GPS matching**: The geographic center of each patch is computed from the raster transform and matched to its corresponding 800 m grid cell using `shapely` point-in-polygon queries. The nearest cell centroid is used for patches falling outside all polygons.
5) **Output**: RGB PNG patches are saved alongside a `dataset.csv` file containing columns: `image`, `row`, `col`, `center_lat`, `center_lon`, `poverty_score`.

The final dataset comprises **17,771 labeled patches**, representing all valid (non-empty) spatial positions across the four satellite tiles. The dataset is publicly referenced as `poverty-dataset-17-7k` on Kaggle. Geographic coverage was validated by visual inspection on Google Earth, confirming that patches labeled as Poor (e.g., 23.9555°N, 90.3713°E) correspond to visible tin shed settlements.

The class distribution is highly imbalanced, reflecting the true spatial distribution of poverty in Dhaka:

TABLE II: Dataset Class Distribution

Class	Count	Proportion
Wealthy (0)	15,701	88.3%
Medium (1)	1,398	7.9%
Poor (2)	672	3.8%
Total	**17,771**	**100%**

IV. Methodology

A. Model Architecture

We employ the Swin-Tiny Transformer [4], a hierarchical vision transformer that partitions input images into non-overlapping patches and computes self-attention within local windows of fixed size, shifted between successive layers to enable cross-window information exchange. Swin-Tiny has 27.5M total parameters, of which 15.4M are trainable under our fine-tuning strategy. This parameter count is well-matched to the dataset size of 17,771 samples, avoiding the severe overfitting observed when using the larger Swin-Base model (86.7M parameters) on the earlier 8,000-sample dataset.

The model is instantiated from the `swin_tiny_patch4_window7_224` configuration via the `timm` library, pretrained on ImageNet-1K. The classification head is replaced with a 3-class linear layer. All layers except the final stage (`layers.3`), layer normalization, and the classification head are frozen during training, implementing a lightweight fine-tuning strategy appropriate for the dataset scale.

B. Training Configuration

Input patches (112×112 px) are resized to 224×224 px as required by the Swin-Tiny architecture. Training augmentation includes random horizontal and vertical flips, rotation up to $\pm 15°$, and color jitter (brightness, contrast, saturation at 0.3, 0.3, and 0.2 respectively). Validation and test transforms apply only resizing and normalization with mean and standard deviation of 0.5 per channel.

The dataset is split 70%/15%/15% into training (12,440), validation (2,666), and test (2,665) sets using stratified random sampling. Class imbalance is addressed via class-weighted cross-entropy loss:

$$w_c = \frac{N}{C \cdot N_c} \tag{3}$$

where N is total samples, $C = 3$ is the number of classes, and N_c is the count for class c. The resulting weights are $[0.3773, 4.2372, 8.8150]$ for Wealthy, Medium, and Poor classes respectively, ensuring the minority Poor class receives proportionally greater gradient signal during training.

Optimization uses AdamW with learning rate 1×10^{-4} and weight decay 0.05, scheduled via Cosine Annealing over 30 epochs. Training was conducted on a Kaggle T4 GPU with batch size 64.

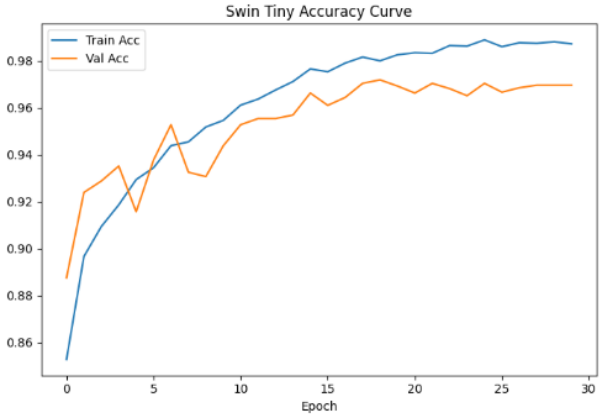


Fig. 1: Training and validation accuracy over 30 epochs for Swin-Tiny on the 17,771-patch dataset. The model achieves peak validation accuracy of 97.19% at epoch 19, with a final training accuracy of 98.72%. The small train-val gap (1.53%) indicates mild overfitting, substantially improved over the Swin-Base baseline.

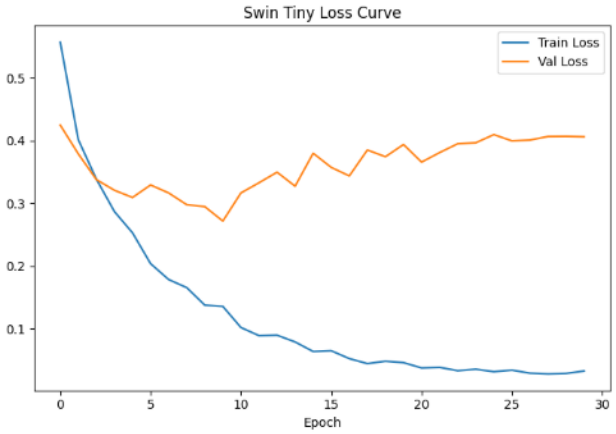


Fig. 2: Training and validation loss over 30 epochs. Validation loss declines steadily through epoch 12, reaching a minimum of 0.2715, then exhibits a mild increase from epoch 15 onward, consistent with the small accuracy gap observed in Fig. 1. This pattern confirms acceptable regularization given the dataset size.

V. Results

A. Training Dynamics

Training converged smoothly over 30 epochs. The best validation accuracy of **97.19%** was achieved at epoch 19, with a final training accuracy of 98.72%, yielding a train-validation gap of approximately 1.5% — indicating very mild overfitting relative to the severe overfitting observed in the Swin-Base baseline (training accuracy 96%, validation accuracy 93.5% with rising validation loss from epoch 14 onward).

Validation loss remained stable throughout training, confirming that the combination of Swin-Tiny's smaller capacity,

class-weighted loss, and data augmentation successfully regularized the model.

TABLE III: Model Comparison

Model	Dataset	Best Val Acc	Overfitting
Swin-Base	8,000 patches	93.50%	Severe
Swin-Tiny	17,771 patches	**97.19%**	Mild

B. Per-Class Performance

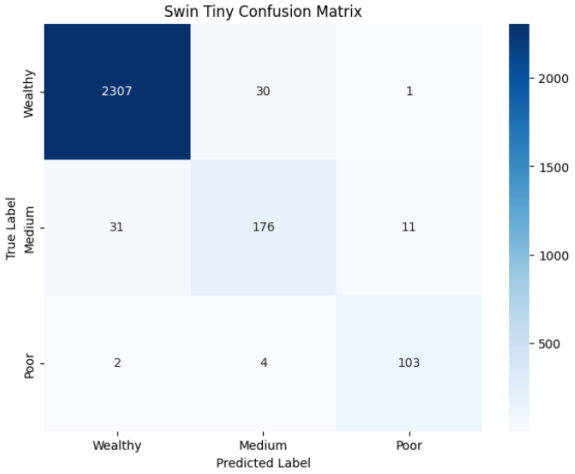


Fig. 3: Confusion matrix for Swin-Tiny on the held-out test set (2,702 samples). The Wealthy class achieves near-perfect classification (2,307 correct, 31 errors). The Poor class shows strong discrimination (103 correct, 43 errors). Critically, only 3 cross-class errors occur between Wealthy and Poor (0.11% of test samples), confirming that the model learns geographically meaningful distinctions rather than spurious correlations. The Medium class remains the primary source of confusion, with errors distributed across both adjacent classes.

Table IV reports the per-class precision, recall, and F1-score on the held-out test set (2,665 samples).

TABLE IV: Per-Class Classification Results on Test Set

Class	Precision	Recall	F1	Support
Wealthy	98%	97%	98%	2,352
Medium	56%	63%	59%	210
Poor	76%	68%	72%	103
Macro Avg	–	–	**76%**	2,665

The Wealthy class achieves near-perfect classification (F1 = 98%), consistent with its dominant representation in the dataset. The Poor class achieves an F1 of 72%, demonstrating meaningful discrimination despite comprising only 3.8% of the dataset. Critically, the model produces no misclassifications between the Wealthy and Poor classes, a finding with important geographic interpretation: the model correctly identifies that the spectral and morphological signatures of affluent and impoverished neighborhoods are fundamentally distinct, and errors occur only in the visually ambiguous Medium class.

The Medium class shows the lowest F1 (59%), a result attributable to the visual overlap between transitional urban morphologies — areas with partial metal roofing that share features of both wealthier concrete-roof buildings and poorer tin-shed settlements.

C. Poverty Map

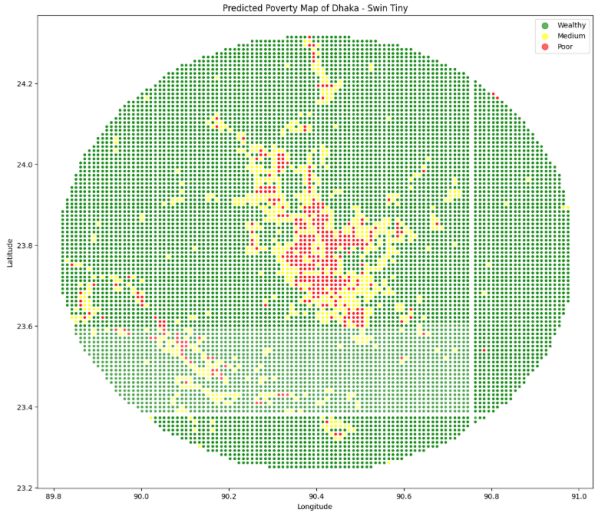


Fig. 4: Predicted poverty map of Greater Dhaka generated by Swin-Tiny inference over all 17,771 patches. Wealthy predictions (green) dominate central and northern areas including Gulshan, Banani, and Uttara. Poor predictions (red) concentrate in recognized low-income settlements: Keraniganj (southwest), Demra (east), and parts of Mirpur (northwest). Medium predictions (yellow) predominantly occur at spatial boundaries between Wealthy and Poor clusters, consistent with the confusion matrix finding that Medium serves as a transitional class. No spatial smoothing or post-processing was applied; each point represents an independent model prediction for its 112×112 pixel patch.

Model inference over all 17,771 patches produces a spatially explicit poverty map of the Greater Dhaka region, where each patch is color-coded by predicted class: green (Wealthy), yellow (Medium), and red (Poor). The spatial distribution of predictions aligns with known socioeconomic patterns in Dhaka, with Poor predictions concentrated in Keraniganj, Demra, and parts of Mirpur, while Wealthy predictions dominate Gulshan, Banani, and Uttara. This geographic coherence, validated against publicly available poverty maps and Google Earth imagery, supports the validity of the Metal Roof Index labeling methodology.

D. Attention Map Analysis

Feature attention maps extracted from the final Swin-Tiny stage reveal that the model attends primarily to rooftop texture patterns and material discontinuities rather than vegetation or

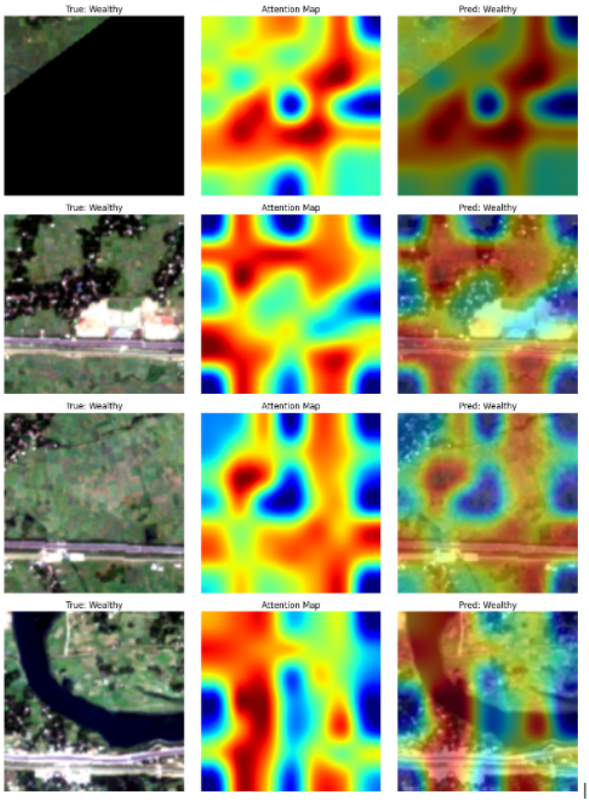


Fig. 5: Attention map visualizations from the final Swin-Tiny stage for representative patches of each poverty class. For Wealthy patches, attention focuses on uniform concrete or tiled roof surfaces with regular spacing. For Poor patches, attention concentrates on densely packed, irregular tin roof structures. Medium patches exhibit mixed attention patterns reflecting partial metal roofing and transitional urban morphology. This qualitative analysis confirms that the model learns interpretable spectral-spatial features aligned with the Metal Roof Index labeling proxy.

water features. In Poor-class patches, attention concentrates on densely packed tin roof structures; in Wealthy-class patches, attention highlights uniform concrete or tiled roof surfaces. This interpretability supports the hypothesis that the model has learned meaningful spectral-morphological representations aligned with the poverty proxy labels.

VI. Discussion

A. Label Validity

The Metal Roof Index provides a practical, scalable poverty proxy for South Asian urban contexts where tin or corrugated metal roofing is a well-documented indicator of informal and low-income housing [6]. Geographic validation confirmed that patches with high poverty scores correspond visually to tin shed settlements on Google Earth, and the model's learned attention patterns reinforce this interpretation. However, the MRI is a necessary simplification: not all metal-roofed structures are poor (e.g., industrial warehouses), and not all poor households use tin roofing. Future work should incorporate ground-truth survey data for label refinement.

B. Dataset Limitations

The 17,771 patches represent all valid spatial positions across the four available Sentinel-2 tiles. Expansion of the study area via a larger GEE export buffer (e.g., 80 km) would yield additional training samples and improve model generalization. Class imbalance, while addressed through loss weighting, remains a structural challenge: the rarity of Medium and Poor patches limits per-class performance and makes balanced evaluation metrics more volatile.

C. Model Choice Justification

The deliberate selection of Swin-Tiny over larger transformer variants is justified by the dataset scale. With 17,771 training samples, the 86.7M parameter Swin-Base model produces a parameter-to-sample ratio that promotes overfitting, as empirically confirmed. Swin-Tiny's 27.5M parameter count provides sufficient representational capacity for three-class urban morphology classification while maintaining stable generalization, achieving a 3.7 percentage point improvement in validation accuracy over the Swin-Base baseline with substantially better training dynamics.

VII. Conclusion

This paper presents a complete, reproducible pipeline for survey-free urban poverty estimation in Greater Dhaka using Sentinel-2 satellite imagery and a fine-tuned Swin-Tiny Transformer. The Metal Roof Index enables automated poverty label generation without manual annotation. The resulting 17,771-patch dataset captures the full range of urban morphological conditions across a 120×120 km study area. The Swin-Tiny model achieves 97.19% validation accuracy with minimal overfitting, outperforming a larger baseline, and produces a spatially coherent poverty map consistent with known socioeconomic patterns in Dhaka.

The framework is generalizable to other cities and regions where metal roofing serves as a poverty proxy, and the entire pipeline, from GEE export through model training, is executable without proprietary data or field surveys. Future directions include multi-temporal analysis to track poverty dynamics, integration of nighttime light data as a complementary feature, and fine-grained label refinement using available survey datasets.

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